Chapter 1

Introduction

Commutative algebra is about studying commutative rings and their modules. The course has a view to algebraic geometry.

Example 1.1 (Examples of Commutative Rings). $\mathbb{Z}$, $k[x_1, ..., x_n]$ for k field. For topological spaces, X we get the commutative ring $C(X) := \{f : X \to \mathbb{C} : f \text{ continous}\}$.

Theorem 1.1. Structure Theorem for Finitely Generated Rings over a PID

Theorem 1.2. $k[x_1, ..., x_n]$ is a UFD

Chapter 2

Basic Definitions

Definition 2.1 (Ring). A **ring** A is a triple $(A, +, \times)$ where A is a set and $+$ and $\times$ are binary operations on A.
1. $(A, +, 0)$ is a abelian group
2. $(A, 1, \times)$ is a monoid.
3. Distributive Laws: $(a + b) \times c = a \times c + b \times c$ and $a \times (b + c) = a \times b + a \times c$

A is a **commutative ring** if and only if $(A, 1, \times)$ is a commutative monoid.

Example 2.1. Examples of noncommutative rings are $M_n(k)$ where k is a field, kG for a nonabelian group G. Examples of commutative rings include $C^\infty(X)$ for X smooth manifold.

If a ring is discussed, then it is commutative from now on unless otherwise stated.

Note on the definition, $1 \in A$ (by definition), though this is not not always required. A zero ring: $0 = 1 \in A$

Exercise 2.1. Show $\forall x \in A$, $0 \cdot x = 0$ and $(-1)x = -x$

Definition 2.2 (Unit). A element $x \in A$ is called a unit (or invertible) if $\exists y \in A$ where $xy = 1 = yx$. An exercise is showing that y is unique. $A^\times$ is the group of units of A.

Definition 2.3 (Zero Divisor). A element $x \in A$ is a **zero divisor** if there exists a nonzero $y \in A$ where $xy = 0$.

Definition 2.4 (Nilpotent). $x \in A$ is **nilpotent** if $\exists n \in \mathbb{N}$ where $x^n = 0 \in A$

Definition 2.5 (Field). A is a field if $1 \neq 0$ and for every $x \neq 0$ in A is a unit.

Definition 2.6 (Integral Domain). A is a **integral domain** or **domain** if $1 \neq 0$ and $\forall x, y \in A$ $xy = 0 \Rightarrow x = 0$ or $y = 0$.

Definition 2.7 (Reduced). A ring A is **reduced** if only nilpotent element in A is 0.

We have the following inclusions:

$$\text{Field} \subset \text{Domain} \subset \text{Reduced}$$

Example 2.2. The zero ring is reduced, but not a domain (or a field). $\mathbb{Z}_p$ is a field if and only if p is prime in $\mathbb{Z}$. $\mathbb{Z}_n$ is not a field if and only if n is not prime. If $\mathbb{Z}_{p_1 \cdots p_n}$ is reduced if $n \geq 0$ and $p_1, ..., p_n$ are distinct.

Definition 2.8 (Subring). A subring of a ring R, $\{0, 1\} \subset A \subset R$ is a subset where A is closed under addition, additive inverses, multiplication, (in R)

Remark 2.1. $\{0\}$ is not a subring of $\mathbb{Z}$

Note in the category of commutative rings, $\mathbb{Z}$ is an initial object.

Definition 2.9 (Left and Right Ideal). An right ideal I in a ring A is a subset where $\forall k \in I$, then $kA \subseteq I$. A left ideal I is subset where $Ak \subset A$. In a commutative ring left and right ideals are equal. Kernel of a ring homomorphism is always a ring, furthermore any ideal is a kernel of a homomorphism

Example 2.3. In a field k, the only ideals are $\{0\}$ and k. If X is a topological space, $Y \subset X$ is a closed (if it isn't closed you can just take closure because of continuity) subspace. Let $I = \ker\left(C(X) \xrightarrow{|_Y} C(Y)\right)$ is a ideal. This is a typical way to get ideals, take a space of functions which is a ring and then restrict on a closed subspace.

Definition 2.10 (The Ideal Generated by a Set). The **ideal generated by $S \subset A$ a ring** denoted (S):
1. The intersection of all ideals containing S.
2. The set of finite linear combinations of elements of S. If $S = \emptyset$. Thus set is the 0 ideal.

For a finite set $S = \{a_1, ..., a_r\} \subset A$ we will used the notation: $(a_1, ..., a_r) = (S)$.

Definition 2.11 (Principal Ideal). An ideal generated by one element.

Note, while the intersection of two ideals is always a ideal, the union is not always a ideal. But the sum of two ideals is a ideal, in fact the smallest ideal containing $I \cup J$.

Definition 2.12 (Quotient Ring). Standard way: R/I is the set of cosets but not ideal Better way: A/I is a ring that come with a ring homomorphism $\varphi : A \twoheadrightarrow A/I$ with kernel I.

Note for the sake of notation: For $a \in A$ we can say $a \in A/I$ to talk about $a + I \in A/I$.

Definition 2.13 (Maximal Ideals). A ideal I is called maximal if A/I is a field.

Definition 2.14 (Prime Ideals). A ideal I is a prime if A/I is a domain

Definition 2.15 (Radical Ideal). A ideal I is radical if A/I is reduced.

Example 2.4 (Maximal, Prime, and Radical Ideas). For a field k, the maximal ideal in $k[x]$ (which is also a PID) is (f) where f an monic irreducible polynomial.

Monic means leading coeefficent is 1. In any **domain** R, $f \in R^\times$ and for $g, h \in R$ $f = gh \Rightarrow$ g or h is a unit for R.

Prime ideals of $k[x]$ are the maximal ideal and 0. Finally the radical ideals of $k[x]$ are (0) and $(f_1, ..., f_r)$ for $1 \leq r < \infty$

Example 2.5 (Prime ideals in a UFD). $\mathbb{Z}[x]$ is a unique factorization domain and not a pid. Consider $(0), (7), (x), (7, x) \subset \mathbb{Z}[x]$. Only $(7, x)$ is maximal.

Example 2.6. Let S be a set, k a field, and A a ring of functiosns $S \to k$ that contains all constant function. Fix $s_0 \in s$

In "good cases", in different points in S give different maximal ideals

Example 2.7. If X a compact Hausdorff space, or a metrizable topological space, by the Tietze Extension Theoerm for any closed subset $Y \subset X$ for any continous $f : Y \to \mathbb{R}$ there is a Extension $X \to \mathbb{R}$. Basically you can seperate points hence the ideals generated are different.

Chapter 3

Spectrum of Ring

Definition 3.1 (Spectrum of a ring). For a commutative ring A,

$$\mathrm{Spec}(A) := \{\text{prime ideals in A}\}$$

Think of elements of A as functions on the space $\mathrm{Spec}(A)$. So A turns into the ring of functions on $\mathrm{Spec}(A)$. For each $\mathfrak{p} \in \mathrm{Spec}(A)$, there is a homomorphism

$$A \to A/\mathfrak{p} \to \mathrm{Frac}(A/\mathfrak{p}).$$

For a function $f \in A$, the value of f at $\mathfrak{p}$ is the image of f in this homomorphism.

Why are we taking the fraction field of a $A/\mathfrak{p}$?

3.1 Why only prime ideals?

We want to take values in a field k. $A \overset{\text{hom}}{\to} k$ maps A to a subring of k, which is a domain, hence the map has a prime ideal as a kernel.

Example 3.1 (Spectrum of Basic Rings).
1. $\mathrm{Spec}\, 0 = \emptyset$
2. If $A \neq 0$, then $\mathrm{Spec}\, A \neq \emptyset$ since A must have a maximal ideal hence a prime ideal.

3.2 Zariski Topology

We will define a topology on $\mathrm{Spec}(A)$.

Definition 3.2 (Zariski Topology of $\mathrm{Spec}(A)$). For a ideal $I \subseteq A$, denote

$$V(I) := \{p \in \mathrm{Spec}(A) : I \subseteq p\}.$$

A subset of $\mathrm{Spec}(A)$ is closed if it is equal to $V(I)$ for some $I \subseteq A$ ideal. A subset of $\mathrm{Spec}(A)$ is open if its the complement of $V(I)$ for some $I \subseteq A$ ideal.

Why does $V(I)$ consist of ideals containing I rather being contained in I? Interpreting elements of A as functions on

$$V(I) := \{\text{pts where all functions in I vanish}\}$$

3.2.1 Nonstandard Notion:

For elements $f_1, ..., f_r \in A$ write $\{f_1 = 0, ..., f_r = 0\} \subseteq \mathrm{Spec}(A)$ for $V((f_1, ..., f_r))$

Theorem 3.3. Show that the closed and open sets defined before form a Topology.

Example 3.2 (Some more Spec).

1. $\text{Spec}(\mathbb{Q}) = \{(0)\}$
2. $\text{Spec}(\mathbb{R}) = \{(0)\}$
3. $\text{Spec}(\mathbb{Z}) = \{(0)\} \cup \{(p) : p \text{ prime}\}$.

$$\{10 = 0\} = \{(2), (5)\}$$

3. $\text{Spec}[\mathbb{C}[x]]$: $\mathbb{C}[x]$ is a PID, and thus we know how the prime ideals look. The prime ideal looks (0) or (f) where $f \in \mathbb{C}[x]$ monic irreducible polynomial. Since $\mathbb{C}$ is algebraic closed, we know that f has degree greater than 1 it can be factored. Thus f must be degree 1 polynomial. $\mathbb{C}[x]/(0)$ is a domain but not a field. $\mathbb{C}[x]/(x-b) \cong \mathbb{C}$. $\text{Spec}(\mathbb{C}[x])$ looks like $\mathbb{C}$ with a extra point (0). The closed subsets of X are the subsets

$$V(I) := \{\mathfrak{p} \in \text{Spec}(\mathbb{C}[x]) : I \subseteq \mathfrak{p}\}.$$

Note that $\mathbb{C}[x]$ is a PID, thus any ideal is $(0), (1), ((x-b_1)^{n_1}...(x-b_n)^{n_r})$ where $b_i \in \mathbb{C}$ and $r \geq 1$.

$$V((0)) = \{0 = 0\} := \text{Spec}(C[[x]])$$
$$V((1)) = \{1 = 0\} = \emptyset$$
$$V((x-b_1)^{n_1}...(x-b_n)^{r_n}) = \{(x-b_1)^{n_1}...(x-b_n)^{n_r} = 0\} = \{(x-b_1), ..., (x-b_n)\}$$

Thus the closed subsets of $\text{Spec}(\mathbb{C})$ looks like finite sets of $\mathbb{C}$ portion of $\text{Spec}[\mathbb{C}]$, empty set, and the whole set. Thus as a topological space, $\text{Spec}(\mathbb{C})$ looks exactly like $\mathbb{C}$ plus a extra point. (0) is called a generic point because the closure of the point is the whole space.

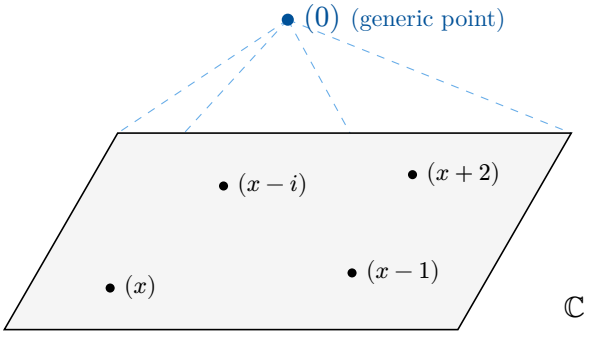


Figure 1: $\text{Spec}(\mathbb{C}[x])$: closed points $(x-b)$ for $b \in \mathbb{C}$, plus the generic point (0), whose closure is all of $\text{Spec}(\mathbb{C}[x])$.

Theorem 3.4 (Nilradical is intersection of all prime ideals). The nilradical, $\mathfrak{N}$, is the set of nilpotent elements of a ring.

$$\mathfrak{N} = \bigcap_{\mathfrak{p} \in \text{Spec}(A)} \mathfrak{p}$$

Proof:

1. $\mathfrak{R} \subseteq \bigcap_{\mathfrak{p} \in \operatorname{Spec}(A)} \mathfrak{p}$: This is trivial because $x \in \mathfrak{R} \Rightarrow x^n = 0$ and thus $x = 0 \in A/\mathfrak{p}$ because $A/\mathfrak{p}$ is a domain, hence $x \in \mathfrak{p}$.
2. $\mathfrak{R} \supseteq \cap_{\mathfrak{p} \in \operatorname{Spec}(A)} \mathfrak{p}$: We will go for a contrapositive. Suppose x is not nilpotent. Then you want $\mathfrak{p} \in \operatorname{Spec}(A)$ such that $x \notin \mathfrak{p}$. Consider the poset of $P = \{I \subseteq \operatorname{Spec}(A) : x^{\mathbb{N}} \notin I\}$ with the comparison relation given by $\subseteq$. Suppose $S = \emptyset$ empty chain, note that $P \neq \emptyset$ because $(0) \in P$. If S is a nonempty chain in P, let

$$J = \bigcup_{I \in S} I \subseteq A$$

is a ideal. Since no interim ideal contains any positive power of x, J doesn't contains any positive power of x. By Zorn's lemma there exists a maximal element I. Clearly $1 \notin I$, because $x \notin I$. Suppose $y \in R - I$, $z \in R - I$ and $yz \in I$. Thus $I \subset I + (y)$ and $I \subset I + (z)$. So $I + (z), I + (y)$

□

Definition 3.5 (Radical Ideal). A ideal I is **radical** if R/I has no nilpotent elements.

Definition 3.6 (Radical of a Ideal). For a ideal I in a ring R, the **radical** of I is

$$r(I) := \{x \in R : (\exists n \geq 1) x^n \in I\} = [R \to R/I]^{-1}(\mathfrak{R})$$

where $[R \to R/I]^{-1}(\mathfrak{R})$ is the inverse image of the nilradical of R/I. Equivalently $r(I)$, is the smallest radical ideal containing I.

Corollary 3.7. For any ideal $I \subseteq R$, $r(I)$ is the intersection of the prime ideals that contain I.

Theorem 3.8. For any ring R and ideals $I, J \subseteq R$. $V(I) = V(J) \Leftrightarrow r(I) = r(J)$.

Chapter 4

Module

Definition 4.1 (Left Module). A **Left Module** over a ring A (not necessarily commutative) is a abelian group M with a action of the ring A such that:
1. $1 * m = m$ for all $m \in M$
2. $(ab) * m = a(b * m)$
3. $(a + b) * m = a * m + b * m$

Remark 4.1. A right A-module M, written $m * a \in M$ for $m \in M$ and $a \in A$. They are not always equal to to the left A-module. For a commutative ring left and right modules are the same.

Example 4.1. For a field k, a k-vector space is a k-module. For a ring A and $n \geq 0$, A^n is a A-modules, called the free module. A $\mathbb{Z}$-module is equivalent to a abelian group. An ideal I in a commutative ring determines two obvious modules: I itself, A/I. In a more general setting, a left ideal is a left submodule of A, a right ideal is a right submodule. For k a field, a $k[x]$-module is a k-vector space M together with a k-linear map from $M \to M$ (action of x). More generally, a $k[x_1, ..., x_n]$ is a k vectorspace with n maps $x_1 : M \to M$, ... $x_n : M \to M$ such that the maps commute. If E is a real vector bundle over a topological space X, then the set M of continuous sections of E is a module over the $C(X)$.

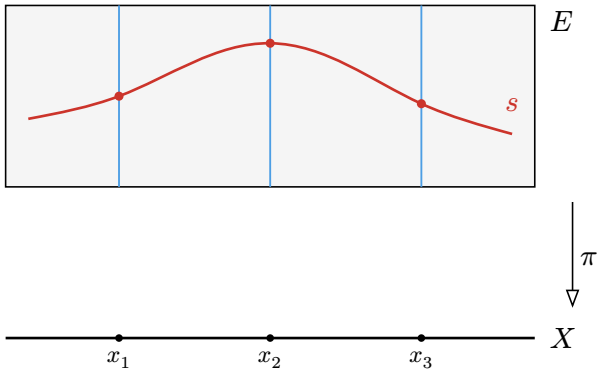


Figure 2: A real vector bundle $\pi : E \to X$. A section s picks one point $s(x)$ in each fiber E_x; the continuous sections form a $C(X)$-module.

When seeing the theorem about modules , consider what it tells you about ideals, by looking at A/I.

The category of A modules is a additive category, unlike (say) the category of commutative rings or commutative A-algebra. Thus A-modules form a nicer simpler category, and studying

A-modules is nicer than studying A. For a commutative ring, $\mathrm{Hom}_A(M, N)$ is a A-module (this is not true for a generic ring).

Definition 4.2. For a A linear map φ:

$$\ker(\varphi) := \{x \in M : \varphi(x) = 0\}$$
$$\mathrm{im}(\varphi) := \{f(m) : m \in M\}$$
$$\mathrm{coker}(f) = N/\mathrm{im}(\varphi)$$

Theorem 4.3 (Universal Property of Quotient Module). Given M a A-module and N a A-submodule, we get a quotient module M/N with a A linear map $\varphi : N \to M/N$ with $\ker \varphi = N$. Equivalently, for any module Q, there is a 1 to 1 correspondence between A-linear maps $N/M \to Q$ and A linear maps $N \to Q$ with kernel which contains M.

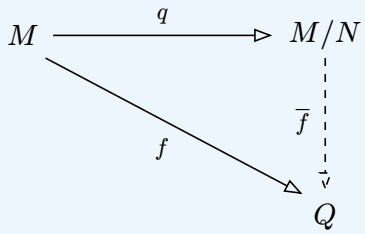


Figure 3: Universal property of M/N: any $f : M \to Q$ with $N \subseteq \ker f$ factors uniquely through the quotient.

Theorem 4.4 (Universal Property of Free Module). For a commutative ring A and any $n \geq 0$, there is a 1 to 1 correspondence from A-linear maps $A^n \to M$ with n-tuples of elements of M.

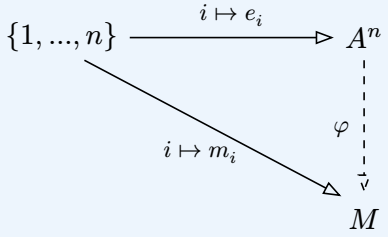


Figure 4: Universal property of A^n: any choice $e_i \mapsto m_i$ of n elements of M extends uniquely to an A-linear map φ.

For any ring homomorphism $A \xrightarrow{f} B$, we get a continuous map

$$\begin{aligned} \mathrm{Spec}\, B &\xrightarrow{f^*} \mathrm{Spec}\, A \\ \mathfrak{p} &\to f^{-1}(\mathfrak{p}). \end{aligned}$$

Thus Spec is a contravariant functor.

Theorem 4.5. For any ideal I in a commutative ring, the ring homomorphism $A \to A/I$ determines a continuous map

$$\operatorname{Spec}(A/I) \to \operatorname{Spec}(A)$$

is a homomorphism into a closed subset of $\operatorname{Spec}(A)$.

Proof: Follow your nose. □

Example 4.2 (What is $\operatorname{Spec}(Z/10)$?). By the theorem, the set is homomorphic to $\{10 = 0\} \subseteq \operatorname{Spec}(\mathbb{Z})$ in otherwords $\{(2), (10)\}$ with the discrete topology.

Lemma 4.6. Let R be a domain. Then $\overline{\{(0)\}} = \operatorname{Spec}(R)$.

Proof: Let S, be a closed subset which contains (0). $\exists I \subseteq R$ such that $S = V(I)$. So $I \subset (0) \Rightarrow I = (0)$. Then $S = V(I) = \operatorname{Spec} R$. □

Corollary 4.7. Let R be any commutative ring, and $\mathfrak{p} \in \operatorname{Spec}(R)$ a point. Then the closure of this point is $V(\mathfrak{p})$.

Proof: $\operatorname{Spec}(A/\mathfrak{p})$ is homomorphic to a closed subset $\operatorname{Spec}(A)$. $V(\mathfrak{p})$. But $(0) \in A/\mathfrak{p}$ comes from $\mathfrak{p}$. By 4.6 we can conclude the result, because $\overline{\{(\mathfrak{p})\}} \cong \overline{\{(0)\}} = \operatorname{Spec}(A/\mathfrak{p}) \cong V(I)$. □

Lemma 4.8. For any ring R, $\operatorname{Max} R \coloneqq \{\text{maximal ideals of } R\}$ is the set of closed points

Proof: Given a maximal ideal $\mathfrak{m} \in \operatorname{Spec} R$. $\overline{\{\mathfrak{m}\}} = V(\mathfrak{m}) = \{\mathfrak{p} \in \operatorname{Spec} R : \mathfrak{m} \subset \mathfrak{p}\} = \{\mathfrak{m}\}$. If $\mathfrak{p} \in \operatorname{Spec}(R)$ is closed, then $V(\mathfrak{p}) = \{\mathfrak{q} \in \operatorname{Spec}(R) : \mathfrak{p} \subseteq \mathfrak{q}\} = \{\mathfrak{p}\}$ hence $\mathfrak{p}$ is not contained in any other prime ideal, hence maximal ideal and must be maximal. □

Definition 4.9 (Irreducible Subspaces). A topological space X is irreducible if $X \neq \emptyset$ and you can't write $X = X_1 \cup X_2$ proper and X_1 and X_2 closed in X.

4.1 Operations on Modules.

Definition 4.10. For a subset S for a R-module M, the submodule generated by S is the intersection of all submodules containing.

Definition 4.11. Biliear Map

Definition 4.12. Tensor product

4.2 Exact Sequence Bullshit

Definition 4.13 (Exact Sequence). An **exact sequence** of R-modules (for non-commutative ring use left module) is a sequence of R-modules $\{M_n\}_{n\in\mathbb{Z}}$ with maps $\{\varphi_n : M_n \to M_{n+1}\}_{n\in\mathbb{Z}}$ such that

$$\ker(\varphi_n) = \mathrm{im}(\varphi_{n-1}).$$

Note this sequence my truncate, ie you may have $\varphi_n = 0$ for $n \geq N$ or $n \leq N$ for $N \in \mathbb{Z}$.

$$\ldots \longrightarrow M_{n-1} \xrightarrow{\varphi_{n-1}} M_n \xrightarrow{\varphi_n} M_{n+1} \longrightarrow \ldots$$

Definition 4.14 (Short Exact Sequence (SES)). A short exact sequence is an exact sequence of the form

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0.$$

4.3 Presentation of Modules

For a R module M, let $(m_i)_{i\in I}$ be a set of generators for M (as a R −module). There exists a surjective map $\varphi : R^{\oplus I} \twoheadrightarrow M$. Let $K = \ker(\varphi)$, an R −submodule of $R^{\oplus I}$. Thus we have an exact sequence

$$K \hookrightarrow R^{\oplus I} \twoheadrightarrow M \to 0.$$

Say that $(u_j)_{j\in J}$ is a set of generators for K, thus we have an exact sequence

$$R^{\oplus J} \cong K \hookrightarrow R^{\oplus I} \to M \to 0$$

4.4 Projective Module

Definition 4.15 (Projective). A R-module M is **Projective** if it's a summand of a free R-module. That is that there exists a set I such $\exists R$-module N such that $R^{\oplus I} \cong M \oplus N$.

Theorem 4.16. Let M be a module over a ring R. The following are equivalent:

1. M is projective
2. Every short exact sequence of R-modules
$$0 \to A \to B \to M \to 0$$
3. Every SES $0 \to A \to B \to C \to 0$ every R-linear map $M \to C$ lifts to B.

Proof:

2. A splitting of a SES is a map $M \to B$ such that $M \to B \to M$ is 1_M. This implies $B = A \oplus M$.
3. A lift of $M \to C$ to $M \to B$ is a R linear map such that the obvious commutative diagram commutes.

□

Example 4.3.

1. $\mathbb{Z}/2\mathbb{Z}$ is not projective as a $\mathbb{Z}$ module. If it was a free module would have torsion.

4.5 Tensor Product

Definition 4.17. Tensor Product

Remark 4.2. Every element of a tensor product module, can be written as a finite sum of decomposable elements

$$\sum_{k=1}^{n} r_k(a_k \otimes b_k)$$

where $r_k \in R$, $a_k \in A$, $b_k \in B$. A decomposable element is $a \otimes b$ where $a \in A$ and $b \in B$. In general, not every element of the tensor product is decomposable.

It isn't clear how to check when two arbitrary finite sums of decomposable elements are equal. For example, consider

$$1 \otimes 1 \in \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/2$$

is equal to 0.

$$\begin{aligned} 1 \otimes 1 &= \left(2\left(\frac{1}{2}\right)\right) \otimes 1 \\ &= 2\left(\frac{1}{2} \otimes 1\right) \\ &= \frac{1}{2} \otimes 2 \\ &= \frac{1}{2} \otimes 0 = 0 \end{aligned}$$

For a non commutative ring R, can define $A \otimes_R B$ for a A a right R-module and B a left R-module. $(ar) \otimes b = a \otimes (rb)$. Broadly we want the "letters to be in the same order". In general $A \otimes_R B$ is a abelian group, but not an R-module. If R is an S −algebra, then $A \otimes_R B$ is at least a S −module.

Exercise 4.1 (Functoriality of Tensor). Tensor product of R module is a functor in each factor. That is given a R-linear map $M_1 \to M_2$ and an R module there is a natural map

$$M_1 \otimes N \to M_2 \otimes N$$

Lemma 4.18. For any commutative ring R A, B, C, there are natural isomorphisms. All tensor products are over R.

$$R \otimes_R A \cong A$$
$$A \otimes B \cong B \otimes A$$
$$(A \otimes B) \otimes C \cong A \otimes (B \otimes C)$$
$$(A \oplus B) \otimes C \cong A \otimes C \oplus B \otimes C$$

The lemma implies that if M and N are free R-module of ranks a and b, then $R^{\oplus a} \otimes R^{\oplus b} \cong R^{\oplus ab}$.

Lemma 4.19. Let $A \to B \to C \to 0$ be a exact sequence of R-module, M an R module Then

$$A \otimes_R M \to B \otimes_R M \to C \otimes_R M \to 0$$

is exact.

Example 4.4. For any element $f \in R$ and a R-module M. Say we have $R \xrightarrow{\times f} R \to R/(f) \to 0$. Lemma gives us

$$M \xrightarrow{\times f} M \to R/(f) \otimes_R M \to 0$$

we know that

$$R/(f) \otimes_R M \cong M/fM$$

Remark 4.3. For a PID R, an R −module is projective $\Leftrightarrow$ its free. By the structure theorem of finite generated Modules over a PID:

$$M \cong R^{\oplus a} \oplus \left(\oplus R/\left(p_i^{m_i}\right)\right)$$

where p_i are irreducible and $m_i \geq 1$.

Example 4.5. $\mathbb{Q}$ is not a projective $\mathbb{Z}$-module, since $\mathbb{Z}$ is a PID and $\mathbb{Q}$ is not free as a $\mathbb{Z}$-module.

4.6 Algebra and Tensor Products

Definition 4.20. A A-algebra B is of a finite type (over A) if B is finitely generated as a A algebra. ($\leftrightarrow B \cong A[x_1, ..., x_n]/I$ for some ideal I).

There is a functor from B-modules -> A modules called "restriction of scalars".

Using $\otimes$ you can also define the functor A-module -> B-modules, "extension of scalars", $M \to M \otimes_A B$ where $b(m \otimes b') = m \otimes bb'$. You show this is well defined using the universal property.

4.7 Flat Modules

Lemma 4.21 (Characterizing Flat Modules). TFAE, for an A module M

1. M is flat as a A-module
2. For every injection $N_1 \to N_2$ of A-modules, $N_1 \otimes_A M \to N_2 \otimes_A M$ is a injection

Chapter 5

Localization

Inverting elements in a (comm) ring A is the same as a working near a point of $\mathrm{Spec}(A)$.

A special case of localization is going from a domain A to the field $\mathrm{Frac}(A)$.

Example 5.1. Consider $\mathbb{C}[x]$. An element $f \in \mathbb{C}[x]$ determines a function $f : \mathbb{C} \to \mathbb{C}$. We write

$$C(x) := \mathrm{Frac}(C[x]) = \{\text{rational functions in one variable} \in \mathbb{C}\}.$$

But $\mathbb{C}(x)$ only defines a function on a open set of $\mathbb{C}$ (specifically $\mathbb{C} - \{\text{finite}\}$). Another example of localizing $C[x]$, is $C[x] \subseteq R \subseteq C(X)$ where R is the ring of rational functions which doesn't explode up to infinity at 0, this is the same as studying $\mathbb{C}$ near the origin.

Definition 5.1. A, a ring, a subset $S \subseteq A$ is called **multiplicatively closed** if S is a submonoid $(A, \cdot)$ (identity, associativity, not necessarily inverses).

Theorem 5.2. For a multiplicatively closed subset $S \subseteq A$, there is a comm ring $A[S^{-1}]$ with a ring homomorphism $f : A \to A[S^{-1}]$ such that

1. $\forall s \in S$, $s \underset{f}{\to} s \in (A[S^{-1}])^\times$
2. $A[S^{-1}]$ is universal for property (1), thus for any comm ring B and a ring hom $g : A \to B$ such that every element of S maps to a unit in B. There exists a unique homomorphism from $A[S^{-1}] \to B$ such that the diagram commutes.

Proof: Show that $at - bs = 0 \Leftrightarrow \frac{a}{s} = \frac{b}{t}$ is not sufficient .

□

Example 5.2. if A is a domain. and $0 \notin S \Rightarrow A \subseteq A[S^{-1}] \subseteq \mathrm{Frac}(A)$. if $0 \in S \Rightarrow A[S^{-1}] = 0$. For any comm ring A. Given that $f \in A$, $S = \{1, f, f^2, ...\}$. We write $A\left[\frac{1}{f}\right] = A[S^{-1}]$. Given any $\mathfrak{p} \subseteq A$, prime ideal. Let $S = A - \mathfrak{p}$ (this is multiplicatively closed if and only $\mathfrak{p}$ is prime). Thus $A_{\mathfrak{p}} = A[A - \mathfrak{p}]$ "localization of A at prime ideal $\mathfrak{p}$".

$\mathbb{Z}\left[\frac{1}{2}\right] = \left\{\frac{n}{2^i} : n \in \mathbb{Z}, i \geq 0\right\}$ is a localization of $\mathbb{Z}$. Compare this to $\mathbb{Z}_{(2)} = \left\{\frac{a}{p} : a \in \mathbb{Z}, b \text{ is odd}\right\}$. Let k be a field.

$$k[x]\left[\frac{1}{x}\right] = \left\{\frac{f(x)}{x^n} : f \in k[x], n \geq 0\right\} = k[x, x^{-1}] \subseteq k(x)$$

$$k[x]_{(x)} = \left\{\frac{f(x)}{g(x)} : f, g \in k[x], g(0) = 0\right\}$$

An element of $k[x, x^{-}1]$ gives a function from $k - \{0\} \to k$. Wheres $f \in k[x]_{(x)}$ gives a function from a open neighborhood of 0 to k.

Theorem 5.3. Let S be a multiplicatively closed subset of a ring A with respect to S. Then the prime ideals of $A[S^{-1}]$ are in one to one correspondence with prime ideals of A that are disjoint to S.

5.1 Local Ring

Definition 5.4. A ring is local if it contains only 1 maximal ideal.

Lemma 5.5. A ring A is local if and only if the nonunits form a ideal

Proof:

□

5.2 Colimits

Many constructions so far (direct sums, quotients, cokernels, $A \otimes_R B$, $A[S^{-1}]$) are instances of a single universal construction: gluing a diagram of objects together along its maps.

Definition 5.6 (Diagram). Let C be a category and S a small category. A **diagram in C of shape S** is a functor $F : S \to C$. Concretely, it is an object $F(i)$ for each $i \in S$ and a morphism $F(\alpha) : F(i) \to F(j)$ for each arrow $\alpha : i \to j$ in S, compatibly with composition and identities.

Definition 5.7 (Cocone, Colimit). A **cocone** on F is an object T of C together with morphisms $g_i : F(i) \to T$ for every $i \in S$ such that $g_j \circ F(\alpha) = g_i$ for every $\alpha : i \to j$.

A **colimit** of F is a cocone (L, ι_i) which is universal: for every cocone (T, g_i) there is a unique morphism $u : L \to T$ with $u \circ \iota_i = g_i$ for all i. We write $L = \operatorname{colim}_{i \in S} F(i)$.

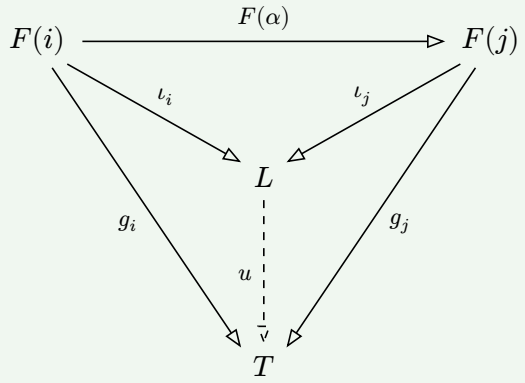


Figure 6: The colimit $L = \operatorname{colim} F$ with legs ι_i. Any other cocone (T, g_i) on F factors uniquely through L via the dashed map u; every triangle commutes.

Remark 5.1. Equivalently, for every object T there is a bijection, natural in T,

$$\mathrm{Hom}_C(\operatorname{colim} F, T) \cong \{\text{cocones on } F \text{ with vertex } T\}.$$

As with every universal property, a colimit (if it exists) is unique up to unique isomorphism. Reversing all arrows gives the dual notion of a **limit** (products, kernels, fibre products, inverse limits).

5.2.1 Examples in A-modules

In the category of A-modules every (small) colimit exists. The basic shapes:

Example 5.3 (Coproduct). If S is discrete (objects only, no non-identity arrows), a cocone is just a family of maps $M_i \to T$, and

$$\operatorname*{colim}_{i \in S} M_i = \bigoplus_{i \in S} M_i$$

with ι_i the canonical inclusions. (In sets this is the disjoint union; in commutative rings the coproduct of B and C is $B \otimes_{\mathbb{Z}} C$.)

Definition 5.8. A **directed set** is a poset P such that $P \neq \emptyset$ such that $\forall a, b \in P$ there exists $c \in P$ such that $c \geq a$ and $c \geq b$

A **directed system** of objects in a category C in a functor $P \to C$ (any poset P gives a category where objects are the elements of P and $\mathrm{Hom}(a, b)$ empty when $a > b$ and a set with 1 element otherwise).

A **direct limit** of a directed system in a category C is a colimit, if it exists.

Example 5.4. In Sets, let $F : P \to \mathbf{Set}$ be a directed system. For the sake of example $P = \mathbb{N}$, $S_1 \xrightarrow{f_1} S_2 \xrightarrow{f_2} \ldots$. The direct limit of this diagram:

$$\varinjlim(S_i, f_i) = \bigsqcup_{a \in P} F(a)/\sim$$

where $x \in F(a)$ and $y \in F(b)$ there exists $c \geq a$ and $c \geq b$ such that x and y map to the same element of $F(x)$.

5.3 Limits

Given any diagram in category C, $F : S \to A$. The limit of F is a object D in C with the morphism $D : A_i$ with formphisms $D \to F$ (1 for every object $a \in S$ such that all diagrams commute) and D is universal with that property.

Definition 5.9 (Cone, Limit). Dually, a **cone** over F is an object T with morphisms $g_i : T \to F(i)$ such that $F(\alpha) \circ g_i = g_j$ for every $\alpha : i \to j$. A **limit** of F is a universal cone (L, π_i): for every cone (T, g_i) there is a unique $u : T \to L$ with $\pi_i \circ u = g_i$ for all i. We write $L = \lim_{i \in S} F(i)$, so $\mathrm{Hom}_C(T, \lim F) \cong \{\text{cones over } F \text{ with vertex } T\}$.

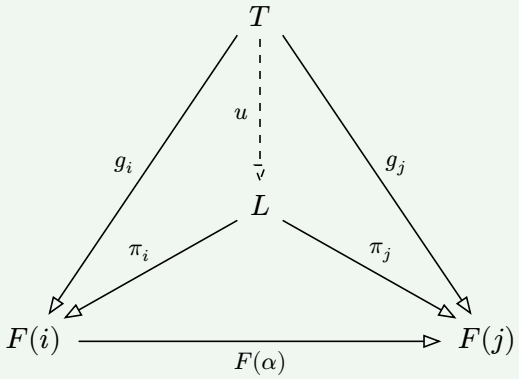


Figure 7: The limit $L = \lim F$ with projections π_i. Any other cone (T, g_i) over F factors uniquely through L via the dashed map u; every triangle commutes.

Example 5.5. In A-modules, limits are computed inside the product:

$$\lim F = \left\{ (x_i) \in \prod_i F(i) : F(\alpha)(x_i) = x_j \text{ for all } \alpha : i \to j \right\}.$$

- Discrete S: the product $\prod_i M_i$.
- $f, g : M \underset{g}{\overset{f}{\rightrightarrows}} N$: the equalizer $\ker(f - g)$; with $g = 0$, the kernel $\ker f$.
- $B \to D \leftarrow C$: the fibre product $B \times_D C = \{(b, c) : f(b) = g(c)\}$.
- $\cdots \to M_3 \to M_2 \to M_1$: the inverse limit, e.g. $\mathbb{Z}_p = \lim \mathbb{Z}/p^n$.

Definition 5.10. A inverse system in a category C is a functor $S^{\mathsf{op}} \to C$ where S is a directed system, op reverses all arrows.

Ex: $A_3 \to A_2 \to A_1$ is a inverse system

Example 5.6. In Sets, or Rmodules, or Comm ring etc.

$$\lim_{\leftarrow} M_\alpha = \left\{ (m_\alpha) \prod_{\alpha \in \text{ ob}(S)} : \text{for every } a \leq b, m_A \to m_b \right\}$$

5.4 Adjoint Functoriality

Given a category A and B, a functor $F : A \to B$ is left adjoint to $G : B \to A$ (we say G is right adjoint to F) if there is a natural bijection of sets $\text{Hom}_B(F(X), Y) \leftrightarrow \text{Hom}_A(X, G(Y))$. for all $X \in A$ and $Y \in B$.

The forgetful functor R-modules to Sets is a right adjoint functor to the free functor from sets to R modules $I \to R^{\otimes I}$.

Chapter 5.4

References